

# SPRINGBOARD VIRTUAL INTERSHIP 6.0

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Domain : Data Visualization

PROJECT : VISUALIZING U.S NATURAL DISASTER DECLARATIONS

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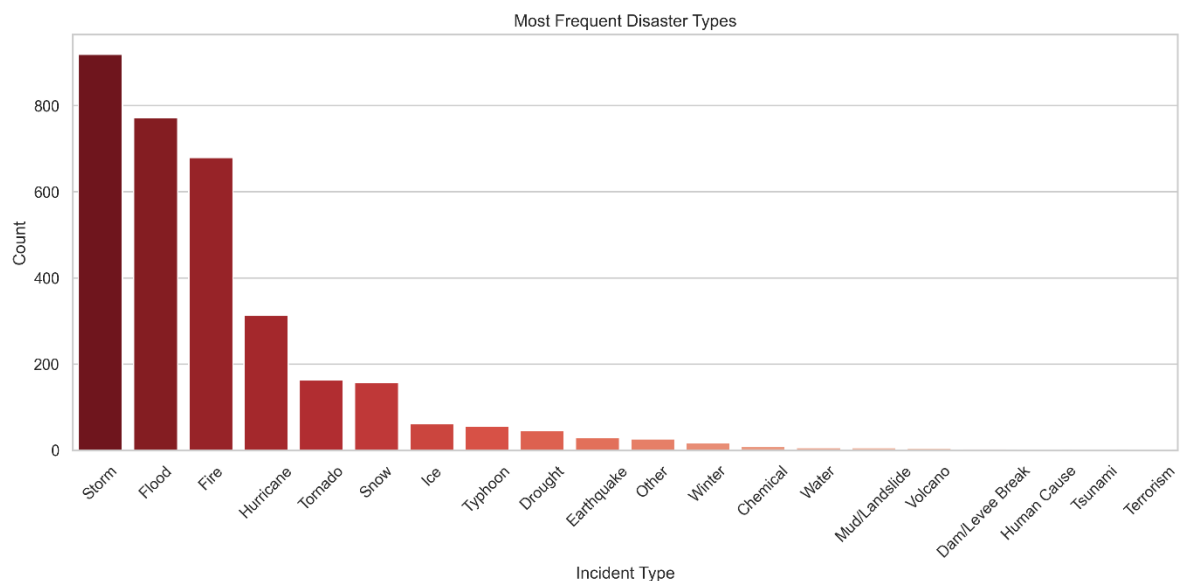
# Milestone 4 – Incident Type Analysis

## Objective:

The objective of Milestone 4 is to perform a detailed analysis of disaster incident types and integrate insights from earlier milestones into one unified dashboard-oriented report. This includes identifying the most frequent disaster categories, understanding how incident types vary across states, analyzing the relationship between incident types and assistance programs (IH and PA), and presenting these findings through clear visualizations to support better disaster preparedness and response planning.

## Visualizations:

### 1.Incident Type Distribution Analysis



### Insights from Incident Type Distribution Analysis

#### Key Findings:

##### 1. Dominant Disaster Categories

- **Storm** is the most frequent incident type with **919 declarations (28.04%)** – representing more than one in four disasters
- **Flood** is second with **772 declarations (23.55%)** – nearly one in four disasters
- **Fire** ranks third with **680 declarations (20.74%)** – one in five disasters
- Together, these three categories account for **72.33%** of all declarations

## 2. Top 5 Concentration

- The top 5 incident types (Storm, Flood, Fire, Hurricane, Tornado) represent **86.85%** of all disaster declarations
- This shows that disaster response planning and resource allocation should prioritize these five categories

## 3. Mid-Tier Threats

- **Hurricane** (313, 9.55%) and **Tornado** (163, 4.97%) are significant but less frequent than weather-related floods and storms
- **Snow** (157, 4.79%) rounds out the top 6, indicating winter weather disasters are also notable

## 4. Long-Tail Distribution

- 11 incident types account for less than 1% frequency each (including Earthquake at 0.88%)
- This "long tail" includes rare events like Drought (1.40%), Typhoon (1.71%), and Ice (1.86%)

## 5. Diversity

- Total of **20 unique incident types** indicates disaster declarations span multiple natural hazard categories
- While diversity exists, the distribution is highly skewed toward dominant types

## 2. Disaster Composition:

Disaster Composition (Treemap)



## Key Composition Insights:

### 1. Three-Tier Hierarchy

- **Major Categories ( $\geq 10\%$ ):** Storm (28.04%) and Flood (23.55%) are the two dominant disaster types
- **Significant Categories (1-10%):** Fire (20.74%), Hurricane (9.55%), Tornado (4.97%), Snow (4.79%)
- **Minor Categories ( $< 1\%$ ):** 11 remaining types including Earthquake, Drought, Typhoon, and others

### 2. Concentration Pattern (Pareto Principle)

- **Top 3 types** comprise **72.33%** of all disasters (Storm, Flood, Fire)
- **Top 5 types** comprise **86.85%** of all disasters
- **Top 10 types** comprise **97.49%** of all disasters
- This follows the 80/20 rule where few categories dominate the landscape

### 3. Weather Dominance

- **Weather-related disasters** (Storm + Flood + Fire + Hurricane + Tornado + Snow + Ice) account for approximately **92%** of all declarations
- Non-weather disasters (Earthquake, Drought, Volcano, Tsunami, etc.) represent only **8%**

### 4. Treemap Visual Representation

- The treemap would show **Storm** occupying the largest rectangular space (28%)
- **Flood** would be the second-largest segment (24%)
- **Fire** would form a substantial block (21%)
- **Hurricane, Tornado, and Snow** would be moderately sized rectangles
- The remaining 11 types would appear as small subdivisions

### 5. Resource Allocation Implications

- **70% of preparedness budget** should address the top 3 disaster types
- **20% of resources** for the next tier (Hurricane, Tornado, Snow)
- **10% reserve** for rare/low-frequency disasters

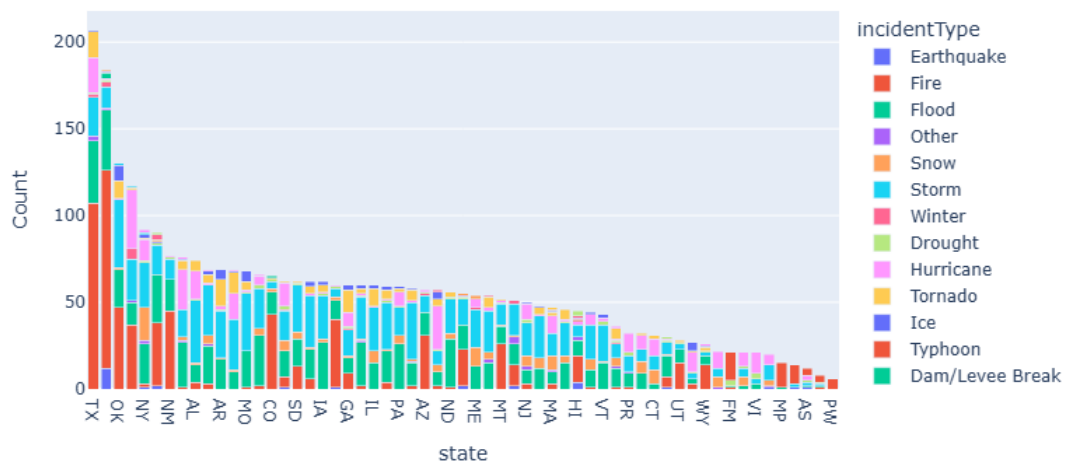
### 6. Long-tail Diversity

- Despite dominance by weather types, the dataset includes **genuine outliers** like Terrorism (2 cases, 0.06%), Tsunami (3 cases), and Human Cause (3 cases)

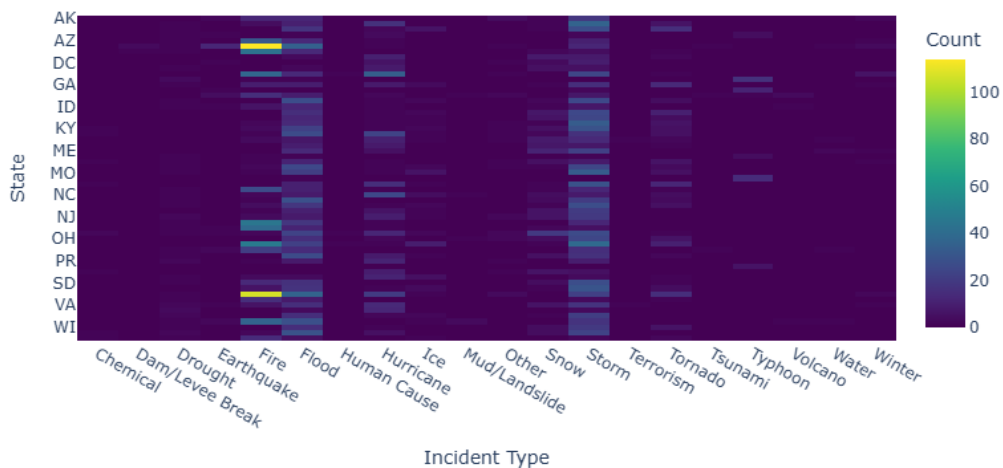
- This tail diversity shows the dataset captures the full spectrum of disaster declarations

### 3.state vs incident type analysis:

Incident Type Contribution within Each State



State vs Incident Type Heatmap



#### 1. Disaster Hotspots

- Texas (207) and California (184) lead; top 10 states = 34% of all declarations
- Texas & California have fire as dominant threat (>50%)

## 2. Regional Patterns

- **West:** Fire-dominant (CA, TX, NM 45-62%)
- **South:** Storms & Hurricanes (FL, AL, LA)
- **Great Plains:** Tornadoes (OK, AR)
- **Appalachia:** Floods (WV, KY)

## 3. Geographic Coverage

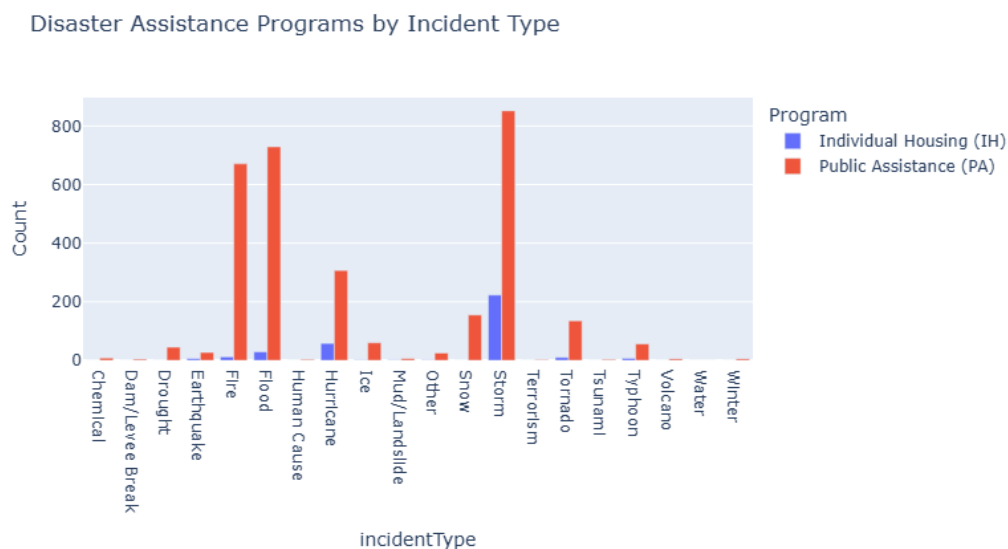
- Storms & Floods: Present in 56-55 states (nationwide threat)
- Fire: 39 states (western concentration)
- Hurricanes: 46 states (coastal focus)
- Tornadoes: 33 states (Midwest/South)

## 4. Diversity

- New York most diverse (12 incident types)
- Small territories least diverse (1-3 types)

5. **Bottom Line:** Disasters are geographically concentrated but incident types vary by region. Focus resources on regional threat profiles.

## 4. Disaster Assistance program analysis:



- **Public Assistance (PA):** 3,098 declarations (94.5%) – dominant program
- **Individual Housing (IH):** 348 declarations (10.6%) – selective use
- **Both Programs:** 302 disasters (9.2%) – co-activation rate
- **Ratio:** PA is ~9x more common than IH

### Top Programs by Incident Type:

| Incident Type | IH %  | PA %  |
|---------------|-------|-------|
| Storm         | 24.3% | 92.7% |
| Hurricane     | 18.5% | 98.1% |
| Flood         | 3.8%  | 94.4% |
| Fire          | 1.8%  | 98.7% |
| Tornado       | 6.1%  | 82.8% |

### Key Insights:

- **All 19 incident types favor PA** – public infrastructure damage dominates response
- **IH rarely triggered alone** – housing assistance is supplementary
- **Housing-critical disasters:** Storms (24% IH) and Earthquakes (21% IH)
- **Infrastructure-critical:** Fire (99% PA), Hurricanes (98% PA), Ice (98% PA)

### Conclusion:

Milestone 4 analysis reveals that US disaster declarations are heavily dominated by three weather-related incident types—Storms (28%), Floods (24%), and Fires (21%)—which together account for 72% of all 3,278 declarations across 59 states. Geographically, disasters concentrate in hotspots like Texas and California, though regional patterns are distinct: the West faces wildfire threats, the South experiences storms and hurricanes, the Great Plains deals with tornadoes, and Appalachia struggles with floods. While disaster declarations are distributed nationwide, Public Assistance programs activate in 94.5% of cases compared to Individual Housing assistance in only 10.6%, indicating that infrastructure damage is the primary concern across all incident types. Strategic disaster preparedness must therefore prioritize weather mitigation in all states while developing region-specific response capabilities, allocating roughly 34% of resources to the top 10 states and maintaining specialized programs for rare but high-impact events.